Deepak Mallampati

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Kent, OH - US-based, F-1 OPT - Open to Berlin/Lisbon relocation (Blue Card)

PROFESSIONAL SUMMARY

Applied AI engineer who ships production ML and deep-learning systems end-to-end - **YOLOv8 real-time computer vision** with ~30% latency cut and ~25% false-positive reduction, transformer-based NLP video summarization with ~85% human alignment, and PyTorch + Hugging Face + FastAPI + Docker deployment discipline (~30% post-deployment defect reduction). Comfortable optimizing models for tight latency and reliability budgets across CV, NLP, and predictive ML. Master's (Kent State). US-based, F-1 OPT.

SKILLS

Deep Learning (PyTorch, Transformers) • Computer Vision (YOLOv8, OpenCV) • NLP & LLM Applications • Model Optimization & Low-Latency Inference • Production ML Pipelines (Docker, FastAPI) • Statistical Modeling & Time-Series

Languages & Frameworks: Python, FastAPI, Flask, SQL (PostgreSQL, SQLite, T-SQL), TypeScript, React, C++ (Arduino), PHP

Deep Learning & ML: PyTorch, Hugging Face Transformers, Diffusers, scikit-learn, XGBoost, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

NLP & LLM: Transformer-based summarization, RAG, embeddings, vector search, structured outputs, prompt engineering, evaluation pipelines

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, RESTful APIs, structured logging, cloud-ready deployment

Domains: Applied AI, Computer Vision, Multimodal Systems, Healthcare AI, Mobile-Style Inference Optimization

PROFESSIONAL EXPERIENCE

Progress Solutions Inc. *Jul 2025 - Present*

Jr. AI/ML Engineer Trainee - USA - Healthcare Technology, AI Consulting, SaaS Healthcare

- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction, care engagement scoring, and support prioritization; improved recall by **15-20%** on high-risk categories while holding precision **>90%** via class weighting, stratified sampling, and threshold calibration.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by **~30%**.
- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval and healthcare documentation search; implemented recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by **~35%** and reduced irrelevant retrieval by **>30%**. Retrieval-grounded response alignment exceeded **90%** in evaluation.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by **~25%**.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; raised dataset reliability above **98%** and cut downstream model instability by **~40%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs.

Suvidha Foundation
Jan 2022 - Mar 2022

Machine Learning Engineer - Hyderabad, India - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions; transcript preprocessing, hierarchical summarization for long-context, and timestamp-aligned clip extraction reduced manual review time by **60-70%** (hours → <5 minutes per video).
- Achieved **~85% alignment** between AI-selected highlights and human-curated key moments.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs **~30%** and raised contextual answer accuracy **>85%** in controlled tests.
- Deployed the stack as a **Flask-based API** with a lightweight web interface for non-technical staff.

Energy Solutions International (Emerson)
Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform (contracts, nominations, allocations, invoicing); reduced query execution time **~20%** and data retrieval latency **~25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates, version-controlled SQL scripts, and stored procedure deployments; reduced deployment errors **>30%** and improved release cycle time **~35-40%** via structured release validation and rollback checkpoints.
- Designed performance dashboards using **SQL Server DMVs and Grafana** to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence **~25%** and improved root-cause resolution speed **~30%**.
- Tuned database maintenance (index rebuilds, statistics updates, partition-aware indexing) and reconciliation queries; improved reconciliation job runtime **~15%** and reduced deadlocks **~15%**.

PROJECTS

Driver Drowsiness Detection System - Real-Time YOLOv8 Fatigue Monitoring

Replaced a two-stage CNN pipeline with a **unified YOLOv8** detection-and-classification model; reduced inference latency **~30%**. LabelImg annotations, augmentation for lighting/head-pose robustness, sliding-window confidence aggregation to suppress blink-driven false positives (**~25% reduction**), adaptive frame skipping, and NMS tuning for stable real-time operation via OpenCV video capture.

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation `shipped, Chrome Web Store`

Full-stack browser extension built independently - Manifest V3, content scripts, service workers, and `captureVisibleTab` for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section capture pipeline handles tall webtoon panels via viewport-slice screenshots with coordinate remapping and dedup. Seven-day translation cache, per-domain selector configs for 29 manga/webtoon sites, privacy policy, and narrowed host permissions.

Video Summarization & Highlight Generation System (Suvidha Foundation)

Transformer-based hierarchical abstractive summarization mapped back to source timestamps for automated clip extraction. **60-70% reduction in manual review time**, **~85% highlight alignment** with human curation, Flask-based API for non-technical users.

Agentic Pixel Character Synthesis & Animation Engine `ongoing`

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. Stable Diffusion fine-tuning + LoRA for identity consistency, ControlNet + OpenPose/MediaPipe for pose conditioning, latent-space consistency for animation smoothness, and an **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

EDUCATION

Master's Degree - Kent State University, Ohio, USA *Aug 2023 - May 2025*